

TIM-150 Project 1 Phase 1 Amended Consulting Report

Optimus Prime Consulting

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1 Introduction

This report will discuss the optimized production allocations for Orange Electronics under the assumption that the Black Friday supply must be met in order to retain B2B and customer relations. Initial solving was done within Excel using the provided solving tools, and while an initial model was formulated, there were issues that prompted the simultaneous development of a python-based OR-Tools model as well.

2 Summary of Findings

2.1 Solution

\$271,319,811 total cost, with **\$11,42,3811** of it being cost-overruns from **express shipping allocated towards Chennai Week 46**, to be deducted from the emergency budget.

Location	Phones	Tablets	Laptops	Shipping
Kaohsiung	0	3094	0	Standard
Chennai	70,000	21,906	1	Standard
Guadalajara	0	0	47,999	Standard
Total	70,000	25,000	48,000	

Table 1: Week 45 Production Targets

Location	Phones	Tablets	Laptops	
Kaohsiung	0	66,906	15,000	Standard
Chennai	165,000	28,094	12,999	Express
Guadalajara	0	0	22,001	Standard
Total	165,000	95,000	50,000	

Table 2: Week 46 Production Targets

Cost Breakdown	
\$251,795,000	Manufacturing
\$19,524,811	Shipping Total
\$8,101,000	Shipping (Standard)
\$11,423,811	Shipping (Express)
\$271,319,811	Total

Table 3: Caption

2.2 Comments on the Solution

While the singular laptop being produced at Chennai for the Week 45 target is odd, manual verification has shown that reallocating the laptop back to Guadalajara would increase final costs by ~\$37 depending on reallocation method.

The distribution of camera modules among the facilities is also crucial, given that Orange Electronics lacks sufficient time to move them between facilities before shipments are due for delivery. Kaohsiung and Guadalajara are constrained by their camera supply in this solution, while Chennai has only 2,000 to spare.

3 Methodology

3.1 Data Cleaning Process and Key Assumptions

When going through the provided stakeholder interviews, we noticed that there were some inconsistencies in the naming conventions, formatting issues, and missing values. For the naming conventions in the datasets, we noticed that there were discrepancies in the names of some of the fields. In the

`Production_Capacity_Data_Oct2025` and

`Shipping_Options_And_Rates` datasets, the fields for the plant names (Kaohsiung, Chennai, and Guadalajara) differ with “Facility” and “Plant”. Similar inconsistencies are seen in the `November_Retail_Orders` dataset in the worksheet tabs, `Week_45_Orders` and `Week_46_Orders`, with “Customer_Name” and “Retailer”, “Delivery_Required” and “Delivery Date”, “Phones_Ordered” and “Phone_Units”, “Tablets” and “Tablet_Demand”, and “Laptop_Orders” and “Laptops_Needed”.

There were also the fields in

`Manufacturing_Cost_Analysis_2025` for the product costs that had varying wording for each of the types. These were some examples, and we standardized all the names listed here and more to make the data easier to read.

The formatting issues we noticed were referring to the cost as USD, but still putting a “\$” in front of the value, and date formatting. The delivery week dates are either written out fully (ex. November 17, 2025) or shortened (ex. Nov 16), and need to be standardized as well for consistency.

The missing values we saw were for the priority ranking of the Customers, how much the “Major cost increase” is for the Charter Flight Threshold, if there are overtime limits, and any values we needed to derive ourselves. We had to make some assumptions here with these values. For the rankings, we prioritized fulfillment for Orange Stores and Enterprise Sales since we need to protect the main retail channel/brand showcase and the customer with the greatest margins. The next highest priority we assumed was Amazon since they require complete fulfillment and comprise 40% of third-party sales. Best Buy and Target are the lowest priority since they allow partial fulfillment. For the Charter Flight Threshold, we chose to assume that a “Major cost increase” wasn’t worth the cost and to make 50,000 units for the carrier capacity a constraint and not exceed it. Regarding the overtime limits for each facility, we made the assumption that there was no upper limit beyond the max weekly shipping capacity and component stock.

3.2 Formulation

We started formulating the model by setting up workbook tabs for the parameters, decision variables, constraints, objective, and solution sheet. After, we filled it out with the information we identified at the problem formulation stage. With the values from the datasets and interviews, we kept the naming conventions consistent throughout.

Solving the problem involved configuring the solver correctly, and iterating on the solver configuration. A well chosen starting value for the solver can greatly reduce solve times and lead to

more reliable convergence onto an optimal solution.

Multiple iterations were required to reformulate the problem into a solvable and actionable state. Abstractions from the precise problem at hand had to be made as well in order to make solving feasible in an Excel environment. A core simplification was assuming cameras would be the main binding variable and that we could assume effectively-infinite volume for other parts at each factory. In alignment with Orange business goals, we also targeted zero late deliveries to ensure customer satisfaction, at the cost of possibly incurring increased overtime and express shipping costs. In order to represent these relationships in our model, we used the following variables and parameters:

Symbol	Description
$X_{p,i,w}$	Quantity of regular-time production units of product i manufactured at plant p for delivery week w .
$X_{p,i,w}^{OT}$	Quantity of overtime production units of product i manufactured at plant p for delivery week w . These units cost 50% more and increase effective capacity by 25%.
$Y_{p,w}$	Binary variable indicating the shipping mode chosen for plant p in week w : $Y_{p,w} = 0$ for standard shipping, $Y_{p,w} = 1$ for express shipping. Only one mode can be used per plant-week.
$S_{c,i,w,p}$	Number of units of product i shipped from plant p to customer c for delivery week w . This variable links plant-level production with customer-level demand and enables on-time delivery enforcement.
$Z_{c,i}$	Binary “all-or-nothing fulfillment indicator” for customer c and product i : $Z_{c,i} = 1$ if the order is fully delivered on time, 0 otherwise. Used primarily for Amazon’s delivery constraint.
$Y_{c,i,w}^{PT}$	Binary variable that activates if customer c’s product i order is delivered late for week w . Used in penalty-cost calculations (5% of contract value per week late).

Table 4: Decision Variables

(See appendix for model parameter table)

Using these variables and parameters (see appendix), we came to the following constraints and objective function:

$$\text{Minimize } Z = \underbrace{\sum_{p,i,w} C_{p,i} X_{p,i,w}}_{\text{Regular production cost}} + \underbrace{\sum_{p,i,w} (1.5 C_{p,i}) X_{p,i,w}^{OT}}_{\text{Overtime cost (50\% premium)}} + \underbrace{\sum_{p,w} (S_p^S (1 - Y_{p,w}) + S_p^E Y_{p,w}) Q_{p,w}}_{\text{Shipping cost}} + \underbrace{\sum_{c,i,w} P_{c,i,w} Y_{c,i,w}^{PT}}_{\text{Penalty cost}}$$

Direct Constraints:

- Camera modules does not exceed 455,000 of total camera modules across all plants

$$\sum_{p, i, w} (X_{p, i, w} + X_{p, i, w}^{OT}) \leq 455,000$$

- Plants cannot use more components than what is available

$$\sum_{i, w} a_i^j (X_{p, i, w} + X_{p, i, w}^{OT}) \leq I_p^j \quad \forall p, j \in \{\text{disp, proc, batt, mem, cam}\}$$

- One shipment per week must be ALL standard or express (pick 1)

$$Y_{p, w} \in \{0, 1\}$$

- Overtime + express shipping cost less than Maximum Emergency Budget Limit

$$\sum_{p, i, w} 0.5 C_{p, i} X_{p, i, w}^{OT} + \sum_{p, w} (S_p^E S_p^S) Y_{p, w} \leq B$$

- Each plant cannot exceed its regular weekly capacity

$$\sum_i r_i X_{p, i, w} \leq \text{Cap}_p$$

- Overtime can only add 25% more capacity cap at most

$$\sum_i r_i X_{p, i, w}^{OT} \leq 0.25 \text{Cap}_p$$

- Total units shipped must meet or exceed each customer's demand

$$\sum_{p, w} X_{p, i, w} + X_{p, i, w}^{OT} \geq d_{c, i}$$

- Production and shipment can't be negative

$$X_{p, i, w}, X_{p, i, w}^{OT}, Q_{p, w} \geq 0$$

- Must satisfy delivery times

$$A_{p, w}^S = \max\{0, (T_w - D_p^S) - T_0\}, \quad A_{p, w}^E = \max\{0, (T_w - D_p^E) - T_0\}$$

$$W_{p, w}^S = \frac{A_{p, w}^S}{7}, \quad W_{p, w}^E = \frac{A_{p, w}^E}{7}$$

$$\sum_i r_i X_{p, i, w} \leq (W_{p, w}^S \text{Cap}_p) (1 - Y_{p, w}) + (W_{p, w}^E \text{Cap}_p) Y_{p, w} \quad \forall p, w$$

$$\sum_i r_i X_{p, i, w}^{OT} \leq 0.25 [(W_{p, w}^S \text{Cap}_p) (1 - Y_{p, w}) + (W_{p, w}^E \text{Cap}_p) Y_{p, w}] \quad \forall p, w$$

Figure 1: Constraints

Validation was an iterative process primarily involving running the solver multiple times and ensuring similar results were obtained each time, in addition to manually editing values to test for trivial failures to find optimality. The Excel solver is known to accumulate floating point error rapidly, especially for complicated problems like this one, so minor deviations from optimal in the final result are to be expected. Testing the Excel solver results against similar solutions from repeated runs showed that the result was within the expected margin of error and was acceptable to issue as a recommendation.

4 Lessons Learned

Over the course of data cleaning, problem formulation, and model development, we learned just how important it is to have clear, well-formatted data. At first, it was a little confusing with all the different field names for the same information. After standardizing them, it was much easier to read and transfer over to the two remaining parts. This is a key part that shouldn't be overlooked and will save time later in the process.

Problem formulation was tedious and took multiple iterations to turn into something usable for the Excel model. Even with the cleaned up data, there were still a lot of moving parts with all the constraints and variables we needed to account for. Our previous assignment helped to give us some experience, but this project was on a different scale. Therefore, once the size of the task is estimated, an appropriate amount of time needs to be allocated to work through and understand what the problem is asking. Rereading and taking notes on the critical information will help you keep track of what needs to be implemented later. Some more advice is to discuss those findings with a group to make sure that everyone is on the same page and that less was missed. When looking at a text or dataset for too long, it becomes easier to overlook what is right in front of you.

Lastly, adjacent to the previous advice, it is important to thoroughly write out your alignment on the problem, identify the decision variables (structural and indicator), formulate the objective function (by defining the parameters, direct relationships, and conditional relationships), and to define the constraints (by defining the parameters, direct constraints, and conditional constraints). With this extensive method, all the components are systematically organized and it is easier to see their relation to each other. Furthermore, it keeps the parameters separate between the objective and the constraints. This splits the mental load of having to figure out what you've already covered/written out.

5 Appendices

5.1 Excel Workbook & Results Overview

[Google Drive Link]

5.2 Model Parameters

Symbol	Description / Source
$p \in \{K, C, G\}$	Plant index: Kaohsiung (Taiwan), Chennai (India), Guadalajara (Mexico).
$i \in \{P, T, L\}$	Product index: Phone (P), Tablet (T), Laptop (L).
$w \in \{45, 46\}$	Delivery week index: Week 45 (Nov 10) and Week 46 (Nov 17).
$c \in \{\text{Orange, Enterprise, Amazon, BestBuy, Target}\}$	Customer index.
$C_{p,i}$	Base production cost of product i at plant p (from Finance).
$1.5C_{p,i}$	Overtime production cost (+50% premium per unit).
r_i	Resource conversion factor: $r_P = 1$, $r_T = 1.5$, $r_L = 2$.
Cap_p	Base weekly capacity: Kaohsiung 52,500; Chennai 80,000; Guadalajara 40,000.
S_p^S	Standard shipping cost per unit from plant p : K \$15; C \$16; G \$5.
S_p^E	Express shipping cost per unit from plant p : K \$60; C \$65; G \$15.
D_p^S, D_p^E	Shipping times (days): K 18/3; C 20/4; G 3/1 (Standard/Express).
$A_{p,w}^S$	Number of production days available for plant p to meet week w using standard shipping.
$A_{p,w}^E$	Number of production days available for plant p to meet week w using express shipping.
$W_{p,w}^S = A_{p,w}^S/7$	Equivalent production weeks available for standard shipments.
$W_{p,w}^E = A_{p,w}^E/7$	Equivalent production weeks available for express shipments.
I_p^{cam}	Available camera modules at each plant; $\sum_p I_p^{\text{cam}} = 455,000$.
a_i^{cam}	Cameras required per product: 1 for all products.
a_i^{mem}	Memory modules required per product: 1 for phones/tablets, 2 for laptops.
$d_{c,i}$	Demand (units) for customer c , product i .
w_c	Due week for each customer: Week 45 for Orange & Enterprise; Week 46 for Amazon, BestBuy, Target.
$P_{c,i,w}$	Contract value or penalty base for customer c , product i , week w .
B	Emergency budget limit = \$15,000,000 (for overtime and express premiums).
e_{OT}	Overtime efficiency = +25% capacity.
$C_{\text{prem}}^E = 0.5C_{p,i}$	Overtime cost premium per unit (counts toward B).
$S_{\text{prem}}^E = S_p^E - S_p^S$	Express shipping cost premium per unit (counts toward B).
$F_{p,w}^S, F_{p,w}^E$	Feasibility flags: 1 if standard/express shipping can meet the delivery week from plant p .
M	Large constant (big-M) used in penalty or feasibility constraints.

Table 5: Parameters

5.3 OR-Tools Model Source

[Google Drive Link]

5.4 What We Fixed

[Google Doc Link]